

MES350 Engineering Statics — Full Course

Worksheet

Pages 1–174 · Prepared for Kitar

Part 1 consists of 57 multiple-choice questions covering every chapter of the course, from General Principles & Units through Friction. Each question has exactly one correct answer among the four choices given. Read each problem carefully, sketch a free-body diagram where appropriate, and select the single best answer. Calculators permitted. No partial credit is awarded for incomplete work on this worksheet — write your final choice clearly. Part 2, beginning after Part 1, contains the answer key with a fully worked explanation for every question.

Part 1 — Worksheet

Ch. 1 — General Principles & Units (Q1–Q6)

1. A car travels at a constant 90 km/h in a straight line. According to Newton's First Law, the net force acting on the car is:
 - (A) Equal to its weight
 - (B) Zero
 - (C) Proportional to its speed
 - (D) Equal to the engine's driving force only
2. A 70 kg person stands on the Moon, where $g \approx 1.62 \text{ m/s}^2$. Compared to standing on Earth, the person's:
 - (A) Mass and weight are both unchanged
 - (B) Mass is unchanged, but weight is smaller
 - (C) Weight is unchanged, but mass is smaller
 - (D) Both mass and weight are smaller
3. You must convert a torque of 25 in·lb into SI units (N·m). Using 1 in = 0.0254 m and 1 lb = 4.448 N, the correct value is closest to:
 - (A) 0.71 N·m

(B) $2.82 \text{ N}\cdot\text{m}$

(C) $11.3 \text{ N}\cdot\text{m}$

(D) $113 \text{ N}\cdot\text{m}$

4. Which of the following best explains why the Mars Climate Orbiter (1998) was lost?
- (A) A structural member failed under an unexpected load
 - (B) One team's software used pound-force while another used Newtons, and the values were never converted
 - (C) The orbiter's mass was miscalculated using the wrong value of g
 - (D) A support was statically indeterminate
5. In an equation for position, $s = vt + \frac{1}{2}at^2$, checking that every term reduces to meters is an application of:
- (A) The parallelogram law
 - (B) Newton's Third Law
 - (C) Dimensional homogeneity
 - (D) The principle of transmissibility
6. Solving a rigid-body equilibrium problem, you find a reaction force $A_x = -40 \text{ N}$ (where you had assumed A_x pointed in the $+x$ direction on your FBD). This means:
- (A) You made an arithmetic error and must redo the problem
 - (B) The reaction is actually 40 N in the $-x$ direction; the magnitude is still correct
 - (C) The structure is not in equilibrium
 - (D) The units must be wrong

Ch. 2 — Force Vectors (Q7–Q14)

7. A ladder leans against a smooth wall and rests on a rough floor. Which statement about the wall reaction is correct?
- (A) The wall provides a normal force only
 - (B) The wall provides both a normal force and a friction force
 - (C) The wall provides a friction force only

(D) The wall provides no reaction at all

8. A force of 200 N acts at 35° above the horizontal. Its vertical (y) component is closest to:

(A) 115 N

(B) 164 N

(C) 200 N

(D) 245 N

9. Vector A has components (6, 8). Its unit vector is:

(A) (6, 8)

(B) (0.6, 0.8)

(C) (3, 4)

(D) (1, 1)

10. Point A is at (1, 2) m and point B is at (4, 6) m. The position vector r_{AB} is:

(A) $3\mathbf{i} + 4\mathbf{j}$ (m)

(B) $5\mathbf{i} + 8\mathbf{j}$ (m)

(C) $-3\mathbf{i} - 4\mathbf{j}$ (m)

(D) $4\mathbf{i} + 6\mathbf{j}$ (m)

11. Two forces $P = 40$ N and $Q = 60$ N act on a bolt with an interior force-triangle angle of 155° between them (Law of Cosines setup). The resultant magnitude is closest to:

(A) 20 N

(B) 50 N

(C) 98 N

(D) 100 N

12. Four forces act on a bolt with x-components +129.9, -27.4, 0, +96.6 (N) and y-components +75.0, +75.2, -110.0, -25.9 (N). The resultant's x-component is:

(A) 14.3 N

(B) 199.1 N

- (C) 253.9 N
 - (D) 199.6 N
13. For the same four forces as the previous question, the resultant's magnitude is closest to:
- (A) 14.3 N
 - (B) 199.1 N
 - (C) 199.6 N
 - (D) 253.9 N
14. Vector addition using the triangle law requires that the second vector be drawn:
- (A) With its tail at the first vector's tail
 - (B) With its head at the first vector's head
 - (C) With its tail at the first vector's head
 - (D) Anywhere, as long as the angle is correct

Ch. 3 — Equilibrium of a Particle (Q15–Q19)

15. A frictionless pulley redirects a rope carrying 250 N of tension. The tension on the other side of the pulley is:
- (A) Less than 250 N
 - (B) Exactly 250 N
 - (C) Greater than 250 N
 - (D) Zero, since the pulley absorbs the force
16. A particle is in equilibrium under exactly three concurrent forces. The minimum number of independent scalar equations needed to solve for two unknowns is:
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 6

17. A 2 kg mass hangs from two symmetric ropes, each at 30° from the horizontal. Using $mg \approx 19.6 \text{ N}$, the tension in each rope is closest to:
- (A) 9.8 N
 - (B) 17.0 N
 - (C) 19.6 N
 - (D) 39.2 N
18. Point A is at (0,0,0) m and point B is at (3,4,0) m. The magnitude of position vector $\mathbf{r}_{AB}$ is:
- (A) 3 m
 - (B) 4 m
 - (C) 5 m
 - (D) 7 m
19. A force's direction cosines satisfy $\cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z =$:
- (A) 0
 - (B) 0.5
 - (C) 1
 - (D) It depends on the force's magnitude

Ch. 4 — Force System Resultants: Moments & Couples (Q20–Q27)

20. A 100 N force acts perpendicular to a wrench, 0.25 m from the pivot. The moment about the pivot is:
- (A) 25 N·m
 - (B) 100 N·m
 - (C) 0.25 N·m
 - (D) 400 N·m
21. In the scalar moment formula $M_O = Fd$, the distance d must be measured as:
- (A) The straight-line distance from O to the force's point of application

- (B) The perpendicular distance from O to the force's line of action
 - (C) The length of the object the force acts on
 - (D) Any convenient distance, since the formula is approximate
22. Varignon's Theorem states that the moment of a resultant force about a point equals:
- (A) The moment of only the largest component force
 - (B) The sum of the moments of the individual component forces
 - (C) The average of the component moments
 - (D) Zero, if the components are perpendicular
23. Two equal, opposite, parallel (non-collinear) forces of 40 N are separated by a perpendicular distance of 0.5 m. This is called a couple, and its moment is:
- (A) 0 N·m, since the forces cancel
 - (B) 20 N·m, the same about every point in space
 - (C) 20 N·m, but only about the midpoint between the forces
 - (D) 80 N·m
24. When a force is moved from its original line of action to a parallel line through a different point O, in order to keep the system equivalent you must add:
- (A) Nothing — moving a force never changes anything
 - (B) A second equal force in the opposite direction only
 - (C) A compensating couple equal to the moment the original force had about O
 - (D) A support reaction
25. A coplanar force system reduces to a resultant force R and a resultant couple MO at point O, with R and MO perpendicular. This system can always be further reduced to:
- (A) A single force, shifted to a new line of action
 - (B) A single couple with no force
 - (C) Two equal and opposite forces
 - (D) It cannot be reduced further

26. The dot product of two perpendicular vectors is:

- (A) Equal to the product of their magnitudes
- (B) Zero
- (C) A vector perpendicular to both
- (D) Undefined

27. For a beam with a 600 N downward resultant located such that $M_A = 1880 \text{ N}\cdot\text{m}$ (clockwise) about point A, the single-force equivalent's distance x from A satisfies $x \cdot R = M_A$. This gives x closest to:

- (A) 1.13 m
- (B) 3.13 m
- (C) 4.8 m
- (D) 6.0 m

Ch. 5 — Equilibrium of a Rigid Body (Q28–Q35)

28. A rigid body has $\sum F = 0$ but $\sum M \neq 0$ about some point. The body is:
- (A) Fully in equilibrium
 - (B) In force equilibrium only; it may still rotate
 - (C) In moment equilibrium only
 - (D) Not a valid physical situation
29. Which 2D support type provides exactly two unknown reaction components (no moment)?
- (A) Roller
 - (B) Pin
 - (C) Fixed (built-in) support
 - (D) Free end
30. A beam has one pin support and one roller support. The total number of unknown reaction components is:
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
31. An 800 N person sits 3 m from a see-saw's pivot. How far from the pivot must a 400 N person sit on the other side to balance it?
- (A) 1.5 m
 - (B) 3 m
 - (C) 6 m
 - (D) 12 m
32. A rigid body in equilibrium under exactly two forces requires those two forces to be:
- (A) Perpendicular to each other

- (B) Equal, opposite, and collinear
 - (C) Equal in magnitude only
 - (D) Concurrent at the body's center of gravity
33. A rigid body in equilibrium under exactly three forces requires the three forces to be:
- (A) All equal in magnitude
 - (B) Either concurrent at one point or all parallel
 - (C) Always perpendicular to one another
 - (D) Applied only at the body's supports
34. A vehicle is on the verge of tipping forward about its front wheels. At that exact instant, the reaction at the rear wheels is:
- (A) At its maximum value
 - (B) Negative
 - (C) Zero
 - (D) Equal to the front wheel reaction
35. For a general 3D rigid body in equilibrium, the total number of independent scalar equilibrium equations available is:
- (A) 2
 - (B) 3
 - (C) 4
 - (D) 6

Ch. 6 — Center of Gravity & Centroids (Q36–Q41)

36. For a triangle with its base on the x-axis and height h , the centroid's height above the base is:
- (A) $h/2$
 - (B) $h/3$
 - (C) $2h/3$

(D) h

37. A flat plate has two lines of symmetry. Its centroid is located:

(A) At the intersection of the two lines of symmetry

(B) Anywhere on either line of symmetry

(C) Cannot be determined without integration

(D) At the plate's edge

38. When computing a composite area's centroid and a circular hole must be removed, the hole's area is treated as:

(A) A positive area, same as the rest of the shape

(B) A negative area

(C) Ignored entirely

(D) Doubled, to account for both its edges

39. An L-shaped bracket is split into two rectangles: Area 1 = 0.5 m^2 at $\bar{x}=0.5 \text{ m}$, Area 2 = 0.125 m^2 at $\bar{x}=0.25 \text{ m}$ (both measured from the same origin O). The composite $\bar{X}$ is closest to:

(A) 0.25 m

(B) 0.375 m

(C) 0.45 m

(D) 0.5 m

40. A distributed load on a beam is triangular in shape. The equivalent concentrated force acts:

(A) At the midpoint of the loaded length

(B) At the centroid of the triangular load diagram

(C) At the point of maximum load intensity

(D) At the beam's support

41. Below the water table, pore water pressure $U = \gamma_w \cdot h$. At the water table itself ($h = 0$), the pore pressure is:

- (A) Maximum
- (B) Zero
- (C) Negative
- (D) Equal to atmospheric pressure

Ch. 7 — Structural Analysis: Trusses, Frames, Machines (Q42–Q48)

42. In a truss, every member is assumed to be a two-force member, meaning loads are applied:
- (A) Anywhere along the member's length
 - (B) Only at the two end joints
 - (C) Only at the member's midpoint
 - (D) Only perpendicular to the member
43. A truss has $M = 9$ members, $R = 3$ reactions, and $J = 6$ joints. Checking $M + R$ against $2J$, this truss is:
- (A) Statically determinate
 - (B) Statically indeterminate
 - (C) Statically unstable
 - (D) Not a valid truss
44. At an unloaded truss joint where exactly two non-collinear members meet and no external load is applied, both members carry:
- (A) Maximum tension
 - (B) Maximum compression
 - (C) Zero force
 - (D) Equal and opposite forces of unknown magnitude
45. The Method of Sections is most efficient when a problem asks for:
- (A) The force in every member of the truss
 - (B) The force in just one or two specific members

- (C) The support reactions only
 - (D) The truss's total weight
46. A frame differs from a truss primarily because a frame contains:
- (A) Only pin connections
 - (B) At least one multi-force member
 - (C) No support reactions
 - (D) Only members in pure compression
47. When two frame members are pin-connected at a shared point C, the pin force that member 1 exerts on member 2 at C, compared to the pin force member 2 exerts on member 1 at C, is:
- (A) Unrelated — each must be found independently
 - (B) Equal in magnitude, same direction
 - (C) Equal in magnitude, opposite direction (same line of action)
 - (D) Always zero
48. For a machine (like a pair of pliers) with input force P at lever arm b and output force Q at lever arm a from the pivot, taking moments gives $Pb = Qa$. If $b > a$, the tool:
- (A) Reduces the output force below the input
 - (B) Amplifies the output force above the input
 - (C) Produces zero output force
 - (D) Cannot be in equilibrium

Ch. 8 — Internal Forces, Stress & Strain, Moment of Inertia (Q49–Q52)

49. A bar carries a tensile force of 1000 lb over a cross-sectional area of 10 in². The stress is:
- (A) 10 lb/in²
 - (B) 100 lb/in²
 - (C) 1000 lb/in²
 - (D) 10,000 lb/in²

50. A bar of original length 10 in. stretches by 0.12 in. under load. Its strain is:
- (A) 0.12 (dimensionless)
 - (B) 0.012 (dimensionless)
 - (C) 1.2 in.
 - (D) 12 in./in.
51. Along a beam segment with zero distributed load ($w = 0$), the shear diagram is:
- (A) A straight sloped line
 - (B) A parabola
 - (C) Flat (constant)
 - (D) Discontinuous
52. The maximum bending moment along a beam occurs where:
- (A) The applied load is largest
 - (B) The shear diagram crosses zero
 - (C) The beam is longest
 - (D) The support reactions are applied

Ch. 9 — Friction (Q53–Q57)

53. Friction force $F_f = \mu F_N$ shows that friction is directly proportional to:
- (A) The contact area between the surfaces
 - (B) The normal force pressing the surfaces together
 - (C) The sliding speed
 - (D) The temperature of the surfaces
54. Comparing static and kinetic friction for the same two surfaces, generally:
- (A) $F_s < F_k$
 - (B) $F_s = F_k$ always
 - (C) $F_s > F_k$

(D) There is no relationship between them

55. A steel block ($\mu_s = 0.74$) rests on a steel surface with normal force 490 N. The maximum static friction force before sliding begins is closest to:

(A) 280 N

(B) 363 N

(C) 490 N

(D) 660 N

56. "Motion impending" means the applied force P equals:

(A) Zero

(B) The kinetic friction force, F_k

(C) The maximum static friction force, F_m

(D) The object's weight

57. Once an object is sliding, the coefficient of kinetic friction μ_k :

(A) Increases with speed

(B) Decreases with speed

(C) Is treated as roughly constant, independent of speed

(D) Becomes equal to μ_s

Part 2 — Answer Sheet

Correct answers are marked in red below each question number, followed by a full worked explanation in red.

Ch. 1 — General Principles & Units

1. Correct answer: **(B) Zero**

Newton's First Law states that a body's velocity stays constant if and only if the net (resultant) force on it is zero — and "constant velocity" explicitly includes cruising at a steady speed in a straight line, not just sitting still. The car here is doing exactly that: 90 km/h, unchanging, straight line. So by the First Law, $\sum F = 0$. Physically, this doesn't mean the engine applies no force — it means the engine's forward push is exactly cancelled by air resistance and rolling friction, so the *net* effect is zero. Option (A) is wrong because weight is a real, nonzero force (it's just balanced by the road's normal force, a separate equilibrium in the vertical direction). Option (C) confuses force with something that scales with speed, which describes drag force in some models, not net force at constant velocity. Option (D) ignores that resistive forces also act on the car; the engine's force alone is not the net force.

2. Correct answer: **(B) Mass is unchanged, but weight is smaller**

Mass is defined as a property of the matter itself — literally "how much stuff" makes up the person — and by definition it does not change no matter where in the universe you take it. Weight, by contrast, is a force: $W = mg$, and g is a local property of whatever gravitational field you're standing in. On Earth $g \approx 9.81 \text{ m/s}^2$, giving $W = 70 \times 9.81 \approx 687 \text{ N}$. On the Moon $g \approx 1.62 \text{ m/s}^2$, giving $W = 70 \times 1.62 \approx 113 \text{ N}$ — noticeably smaller, roughly one-sixth of the Earth value, which matches the everyday fact that astronauts feel "lighter" on the Moon even though their body (their mass) hasn't changed at all. This is precisely the mass-vs-weight distinction the course stresses: never say "this box weighs 5 kg," because kg measures mass, not weight.

3. Correct answer: **(B) 2.82 N·m**

Use the "multiply by 1" conversion-factor method: write each conversion as a fraction that equals 1, with the unit you want to cancel on the bottom. $25 \text{ in} \cdot \text{lb} \times (0.0254 \text{ m} / 1 \text{ in}) \times (4.448 \text{ N} / 1 \text{ lb})$. The "in" units cancel (one in numerator from the original quantity, one in denominator of the first fraction), and the "lb" units cancel the same way, leaving only N·m. Numerically: $25 \times 0.0254 \times 4.448 = 25 \times 0.11298 \approx 2.824 \text{ N} \cdot \text{m}$, which rounds to 2.82 N·m. Notice how far off the wrong-looking answers are: (A) 0.71 N·m is what you'd get if you forgot to convert the pound-force part; (C) and (D) are off by factors of 4 and 40 respectively, which is the kind of error that slips through if you don't carefully track which conversion factor divides and which multiplies.

4. Correct answer: **(B) One team's software used pound-force while another used Newtons, and the values were never converted**

The 1998 Mars Climate Orbiter is the textbook cautionary tale for this exact mistake: NASA's navigation team worked entirely in SI units (Newtons), while spacecraft builder Lockheed Martin's software calculated and output thruster performance data in Imperial pound-force units. Nobody inserted a conversion step between the two systems, so a number that was correct in pound-force was fed directly into equations expecting Newtons — off by a factor of about 4.45. The resulting navigation errors accumulated over the cruise to Mars until the spacecraft entered the Martian atmosphere at the wrong altitude and was destroyed. Options (A) and (C) describe plausible-sounding but historically incorrect causes; (D) is a real engineering concept but unrelated to this specific failure.

5. Correct answer: **(C) Dimensional homogeneity**

Dimensional homogeneity is the rule that every term you're adding together in a physically valid equation must reduce to the same base units — you can't add meters to meters-per-second, for instance, any more than you can add 3 apples to 5 oranges and call it "8" of a single thing. Checking $s = vt + \frac{1}{2}at^2$: s is in meters; vt has units $(\text{m/s})(\text{s}) = \text{m}$; $\frac{1}{2}at^2$ has units $(\text{m/s}^2)(\text{s}^2) = \text{m}$. All three terms reduce to meters, so the equation passes the check — this doesn't prove the equation is physically correct (a dimensionally-consistent equation can still have the wrong coefficient), but it does catch many common algebra mistakes for free, before you even plug in numbers. The parallelogram law (A) and Newton's Third Law (B) are unrelated vector/force concepts; the principle of transmissibility (D) is about sliding a force along its own line of action, also unrelated.

6. Correct answer: **(B) The reaction is actually 40 N in the $-x$ direction; the magnitude is still correct**

When you draw an FBD, you have to guess a direction for each unknown reaction before you know the answer — that's unavoidable, since you don't know the answer yet. If your equilibrium equations then hand back a negative number for that reaction, the mathematics is telling you your guessed direction was backwards, not that you made an arithmetic mistake. The 40 N magnitude itself is exactly as trustworthy as a positive result would have been; you simply flip the arrow on your final answer sketch. Re-doing the whole problem (A) is unnecessary — it would just produce the same negative number again, since the algebra is not the problem. This is a very common point of confusion for students first transitioning into rigid-body equilibrium problems, so the course explicitly calls it out.

Ch. 2 — Force Vectors

7. Correct answer: **(A) The wall provides a normal force only**

In statics, the word "smooth" is a technical term meaning frictionless — it is not a casual description of texture. A frictionless surface, by definition, cannot exert any force component parallel to itself (no friction), so the only thing it can push with is a force perpendicular to its own surface: a normal force. This matters directly for how you draw the ladder's free-body diagram: at the wall contact point you draw exactly one arrow (normal, pointing away from the wall, horizontal if the wall is vertical); you do NOT add a second friction arrow there, even though the ground contact (which is described as rough, $\mu_s=0.40$) does get both a normal force and a friction force. Mixing this up — adding friction at the smooth wall, or forgetting it at the rough ground — is called out explicitly as the most common mistake on this exact problem.

8. Correct answer: **(A) 115 N**

To find the vertical component of a force given its magnitude and the angle it makes with the horizontal, use $F_y = F \sin \theta$ (sine pairs with the angle-from-horizontal convention because the vertical side is "opposite" that angle in the right triangle formed by the force and its components). Here: $F_y = 200 \times \sin(35^\circ) = 200 \times 0.5736 \approx 114.7 \text{ N}$, which rounds to 115 N. A common error is swapping sine and cosine — using $F \cos \theta$ instead — which would give roughly 163.8 N, close to distractor (B) 164 N; that distractor exists specifically to catch that swap. Always re-derive which trig function applies from a quick sketch of the right triangle rather than from memory of "which one goes with y," since the correct pairing depends on whether the angle is measured from the horizontal or the vertical axis.

9. Correct answer: **(B) (0.6, 0.8)**

A unit vector points in the same direction as the original vector but has magnitude exactly 1. To build it, divide every component of the vector by the vector's own magnitude. First find the magnitude: $|A| = \sqrt{6^2 + 8^2} = \sqrt{36+64} = \sqrt{100} = 10$. Then divide each component: $(6/10, 8/10) = (0.6, 0.8)$. As a sanity check, a true unit vector's own magnitude must equal 1: $\sqrt{(0.6)^2 + (0.8)^2} = \sqrt{0.36+0.64} = \sqrt{1} = 1 \checkmark$. Option (A) is simply the original (un-normalized) vector, a common slip when a student forgets the "divide by magnitude" step entirely; (C) and (D) don't correspond to any meaningful operation on this vector.

10. Correct answer: **(A) $3\mathbf{i} + 4\mathbf{j}$ (m)**

The position vector from point A to point B is always found "head minus tail": $\mathbf{r}_{AB} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j}$, where B (the destination, the vector's head) is subtracted-from and A (the starting point, the vector's tail) is what you subtract. Plugging in: $\mathbf{r}_{AB} = (4-1)\mathbf{i} + (6-2)\mathbf{j} = 3\mathbf{i} + 4\mathbf{j} \text{ (m)}$. This literally reads as "to walk from A to B, go 3 m in the +x direction and 4 m in the +y direction." Option (C), $-3\mathbf{i} - 4\mathbf{j}$, is exactly what you'd get if you reversed the subtraction order and accidentally computed $\mathbf{r}_{BA}$ (from B to A) instead of $\mathbf{r}_{AB}$ — a very easy mix-up if you don't carefully track which point is the tail.

11. Correct answer: **(C) 98 N**

When you know the magnitudes of two forces and the angle between them as drawn in the closed force triangle, the Law of Cosines gives the resultant directly: $R^2 = P^2 + Q^2 - 2PQ \cos(B)$, where B is the triangle's interior angle (here 155°). $R^2 = 40^2 + 60^2 - 2(40)(60)\cos(155^\circ) = 1600 + 3600 - 4800 \times (-0.9063) = 5200 + 4350.3 = 9550.3$, so $R = \sqrt{9550.3} \approx 97.7$ N, which rounds to the closest listed option, 98 N. Notice the minus-times-minus in the cosine term: $\cos(155^\circ)$ is negative (since 155° is obtuse), so subtracting a negative actually ADDS to the total — this is why the resultant (97.7 N) ends up larger than a simple average of 40 and 60 might suggest. Getting the sign of that cosine term wrong is the most common error with this formula.

12. Correct answer: **(B) 199.1 N**

When combining several concurrent forces by the component method, you build a table of each force's x- and y-components (with the correct sign for whichever direction it actually points), then sum straight down each column separately — the x-components only interact with other x-components, never with y. Summing the x-column here: $129.9 + (-27.4) + 0 + 96.6 = 199.1$ N. Option (D), 253.9 N, is what you get if you forget the second force's x-component is negative and add all four values as if they were positive — a very common slip when a force points in the "negative" direction on your chosen axes but its magnitude was written down as a plain positive number.

13. Correct answer: **(C) 199.6 N**

First find the resultant's y-component the same way, by summing the y-column: $75.0 + 75.2 + (-110.0) + (-25.9) = 14.3$ N. With both resultant components known ($R_x = 199.1$ N from the previous question, $R_y = 14.3$ N here), the magnitude follows from the Pythagorean theorem, since R_x and R_y are perpendicular: $R = \sqrt{R_x^2 + R_y^2} = \sqrt{(199.1)^2 + (14.3)^2} = \sqrt{39,640.8 + 204.5} = \sqrt{39,845.3} \approx 199.6$ N. Notice how small the y-component's contribution is (14.3 N) compared to the x-component (199.1 N) — this is why the final resultant magnitude (199.6 N) barely differs from R_x alone, and also why the resultant's angle above the x-axis, $\tan^{-1}(14.3/199.1) \approx 4.1^\circ$, is quite shallow.

14. Correct answer: **(C) With its tail at the first vector's head**

The triangle law for vector addition requires drawing the vectors "head to tail": you draw the first vector, then start the second vector's tail at exactly the point where the first vector's head ended — like plotting a walking path, where each new leg of the journey starts where the last one left off. The resultant is then the single vector drawn from the very first tail straight to the very last head, "shortcutting" the whole path. A very common mistake is connecting head-to-head or tail-to-tail instead (options that don't appear correctly here, but are the wrong pairings students often try), which produces a completely different, incorrect resultant. Because vector addition is commutative, it doesn't matter which vector you draw first — $U+V$ and $V+U$ trace different intermediate paths but land on the identical final resultant.

Ch. 3 — Equilibrium of a Particle

15. Correct answer: **(B) Exactly 250 N**

A pulley's entire job, in the idealized frictionless/massless model used throughout this course, is to change the DIRECTION a rope pulls without changing how HARD it pulls. Physically this follows from the pulley itself being in equilibrium: if the tension on one side were different from the tension on the other side, the pulley would experience a net tangential force and either accelerate or require friction to hold it in place — neither of which is assumed here. So whatever tension exists on one side of the rope (250 N) must be identical on the other side. This is a genuinely different idea from a lever or gear, which CAN trade force for distance — a single pulley never does that; only more elaborate block-and-tackle systems with multiple pulleys can multiply force, and that's a separate, more advanced topic.

16. Correct answer: **(B) 2**

A particle, by the course's own definition, is a body whose size and shape are ignored — every force acting on it is treated as passing through the same single point. That means a moment equation taken about that point is automatically satisfied (every force has zero perpendicular distance to its own point of concurrency, so every term in $\sum M$ is zero) — it gives you " $0=0$," true but useless, contributing no new information. That leaves exactly two genuinely independent equations, $\sum F_x=0$ and $\sum F_y=0$, which is why a particle-equilibrium problem can solve for at most two unknowns at a time. This is different from a rigid body (Ch.5), where forces do NOT all act at one point and the moment equation becomes a third, genuinely independent tool.

17. Correct answer: **(C) 19.6 N**

Because both ropes make the identical 30° angle with the horizontal, the setup is left-right symmetric, so by symmetry alone (no algebra needed yet) both tensions must be equal: call them T . The weight to support is $mg = 2 \text{ kg} \times 9.81 \text{ m/s}^2 \approx 19.6 \text{ N}$, acting straight down. Vertical equilibrium requires the two ropes' vertical components to add up to exactly that weight: $T \sin 30^\circ + T \sin 30^\circ = mg \rightarrow 2T \sin 30^\circ = 19.6 \text{ N} \rightarrow 2T(0.5) = 19.6 \rightarrow T = 19.6 \text{ N}$. (The horizontal components of the two ropes are equal and opposite by the same symmetry, so they cancel automatically and don't even need to be checked.) This symmetric shortcut only works because the two angles genuinely match — the course explicitly warns that unequal angles give unequal tensions, requiring the full two-equation method instead.

18. Correct answer: **(C) 5 m**

In 3D (or 2D, as here with $z=0$ for both points), the magnitude of a position vector is found with the 3D distance formula, which is just the Pythagorean theorem extended: $|r_{AB}| = \sqrt{((x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2)} = \sqrt{((3-0)^2 + (4-0)^2 + (0-0)^2)} = \sqrt{(9 + 16 + 0)} = \sqrt{25} = 5 \text{ m}$. You may recognize 3-4-5 as the smallest whole-number right triangle — it's deliberately chosen here so the arithmetic comes out to a clean, memorable answer, but the SAME formula applies for any set of coordinates, whole numbers or not.

19. Correct answer: **(C) 1**

This identity falls directly out of the Pythagorean theorem applied to a vector's components. If a vector F has magnitude $|F|$ and makes angles θ_x , θ_y , θ_z with the three coordinate axes, its components are $F_x = |F|\cos\theta_x$, $F_y = |F|\cos\theta_y$, $F_z = |F|\cos\theta_z$. Since F itself satisfies $|F|^2 = F_x^2 + F_y^2 + F_z^2$ (Pythagoras in 3D), substituting gives $|F|^2 = |F|^2\cos^2\theta_x + |F|^2\cos^2\theta_y + |F|^2\cos^2\theta_z$, and dividing both sides by $|F|^2$ leaves $\cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z = 1$ — true for ANY vector, regardless of its magnitude or specific direction. This is exactly why, if you're given only two of the three direction-cosine angles, you can always solve for the third (up to a sign ambiguity, since cosine squared can't tell you if the missing angle is acute or obtuse).

Ch. 4 — Force System Resultants

20. Correct answer: (A) 25 N·m

The scalar moment formula is $M = F \cdot d$, where d must be the perpendicular (shortest) distance from the moment center to the force's line of action. Because this force is stated to act perpendicular to the wrench, the wrench's own length IS already that perpendicular distance — no trigonometry is needed to convert it. So $M = 100 \text{ N} \times 0.25 \text{ m} = 25 \text{ N}\cdot\text{m}$ directly. If the force had instead been applied at some angle other than 90° to the wrench, you would first need to find the true perpendicular distance (often via $d = L \sin\theta$, where L is the wrench length and θ the angle between the force and the wrench), rather than using the wrench's full length as-is.

21. Correct answer: (B) The perpendicular distance from O to the force's line of action

This is the single most important — and most commonly misapplied — detail in the whole moment chapter: d in $M_O = Fd$ is NOT the distance from O to wherever the force happens to be drawn touching the body. It is the shortest (perpendicular) distance from O to the force's entire line of action, extended infinitely in both directions if necessary, as if the line were a laser beam passing through the force arrow. Using the wrong distance (for example, the straight-line distance to the force's application point instead of the perpendicular distance to its full line) is called out explicitly as the single most common moment-calculation mistake in the course. When the force and the line connecting O to the application point aren't already perpendicular, you generally need Varignon's Theorem or basic trigonometry to find the true perpendicular distance.

22. Correct answer: (B) The sum of the moments of the individual component forces

Varignon's Theorem is what makes the moment calculation practical when a force is applied at an awkward angle: instead of hunting for the true perpendicular distance to the whole tilted force (which can require extra geometry), you split the force into its x - and y -components first, then find the moment of EACH component separately (which is often much easier, since one component may even line up with the position vector and contribute zero moment), and simply add those two moments together. The theorem guarantees this gives the exact same answer as computing the moment of the original, undivided force — it's a computational shortcut, not an approximation.

23. Correct answer: **(B) 20 N·m, the same about every point in space**

A couple is two forces, equal in magnitude, opposite in direction, with parallel (but not identical) lines of action. Its moment is $M = F \cdot d = 40 \text{ N} \times 0.5 \text{ m} = 20 \text{ N} \cdot \text{m}$. What makes a couple special — and different from a single force's moment, which DOES depend on where you measure it from — is that a couple's net force is exactly zero (the two forces cancel), so there's no "leftover force" whose moment could change depending on your reference point; only the pure rotational tendency remains, and that quantity is identical from every vantage point in space. This is why a couple is called a "free vector": you can slide it to any point on the body without changing its physical effect.

24. Correct answer: **(C) A compensating couple equal to the moment the original force had about O**

You genuinely cannot just "pick up" a force and set it down at a different, non-collinear point — doing so would change the body's rotational behavior, since the moment of a force depends on its exact line of action. To move a force from its original line to a new point O while keeping the system's overall effect on the rigid body identical, you must ADD a couple whose moment exactly equals $M_O = r \times F$ — the moment the original force used to produce about O before it was moved. The resulting "force-couple system" (the force, now acting at O, plus this new couple) is physically equivalent to the original single offset force. This is different from the Principle of Transmissibility, which only allows sliding a force ALONG its own existing line of action — no couple needed there, because nothing about the moment changes when you slide along the same line.

25. Correct answer: **(A) A single force, shifted to a new line of action**

Once a force-couple system's resultant force R and resultant couple M_O turn out to be perpendicular to each other (which is automatically guaranteed whenever the original forces were concurrent, coplanar, or all parallel — the three special cases named explicitly in the deck), you can eliminate the separate couple entirely by sliding R to a new, parallel line of action. The perpendicular offset distance for that new line is found by rearranging $M = R \cdot d$ into $d = M_O/R$. This works because a force R combined with a couple M_O , when the two are perpendicular, is exactly equivalent to the same force R alone, just displaced sideways by d — the couple effectively "explains itself away" once you account for the shift.

26. Correct answer: **(B) Zero**

The dot product formula is $P \cdot Q = |P||Q|\cos\theta$, where θ is the angle between the two vectors. For perpendicular vectors, $\theta = 90^\circ$, and $\cos(90^\circ) = 0$ exactly — so no matter how large or small |P| and |Q| are, their dot product is always precisely zero when they're perpendicular. This fact is used constantly as a diagnostic tool: if you compute a dot product and get zero, that's a direct confirmation the two vectors are perpendicular (this is exactly how the deck's cube example, Ex.5(d), proves that force P is perpendicular to diagonal AG — by showing $P \cdot \lambda = 0$).

27. Correct answer: **(B) 3.13 m**

Once you know a system's resultant force R and its resultant moment MA about some reference point A , you can find where along the line of action a single equivalent force must be placed to reproduce that same moment. Setting up $x \cdot R = MA$ (the moment the shifted single force R would create about A , at a distance x away, must equal the already-known moment MA): $x(600 \text{ N}) = 1880 \text{ N} \cdot \text{m} \rightarrow x = 1880/600 \approx 3.133 \text{ m}$, which rounds to 3.13 m. This is exactly the beam example from the study guide (Example 7), where a set of four separate point loads reduces first to a force-couple system at A , and then further reduces to one single 600 N force acting 3.13 m from A — with completely identical effect on the beam as the original four loads combined.

Ch. 5 — Equilibrium of a Rigid Body

28. Correct answer: **(B) In force equilibrium only; it may still rotate**

A rigid body (unlike the particles from Ch.3) needs BOTH $\sum F=0$ AND $\sum M=0$ to be truly in equilibrium — these are two genuinely independent conditions, and satisfying one says nothing about the other. The classic example: a light rod with two equal, opposite forces applied at its two ends, perpendicular to the rod (a couple). The forces cancel perfectly ($\sum F=0$ — force equilibrium holds), but because they act along different, parallel lines rather than the same line, they both contribute a moment in the SAME rotational sense, so $\sum M \neq 0$. The rod will start spinning even though it never starts translating. Checking only the force condition and declaring "equilibrium" is explicitly flagged as the most common mistake when students move from particle problems into rigid-body problems.

29. Correct answer: **(B) Pin**

Each 2D support type is defined by exactly which motions it physically blocks, and it only provides a reaction force/moment in the direction(s) it restrains. A roller blocks motion in only one direction (perpendicular to the rolling surface) — 1 unknown. A pin blocks BOTH horizontal and vertical translation but still allows the connected member to rotate freely about the pin — 2 unknowns (A_x , A_y), but no moment reaction, since rotation isn't restrained. A fixed (built-in/welded) support blocks translation AND rotation entirely — 3 unknowns (A_x , A_y , and a moment MA). So the support providing exactly two unknowns with no moment is specifically the pin.

30. Correct answer: **(C) 3**

Count each support's own unknowns and add them together: a pin contributes 2 (A_x and A_y), and a roller contributes 1 (just the single perpendicular reaction, B_y). Total unknowns = $2 + 1 = 3$. This matches exactly the 3 independent 2D equilibrium equations available for any rigid body ($\sum F_x=0$, $\sum F_y=0$, $\sum M=0$) — 3 equations, 3 unknowns, a perfectly solvable (statically determinate) system. If there had been, say, two pins instead of a pin and a roller, that would be $2+2=4$ unknowns against only 3 equations — statically indeterminate, unsolvable by statics alone.

31. Correct answer: **(C) 6 m**

For a rigid body pivoted at one point (a see-saw), equilibrium requires the clockwise moments about the pivot to exactly balance the counterclockwise moments. Setting the two moments equal: $(800\text{ N})(3\text{ m}) = (400\text{ N})(d) \rightarrow 2400\text{ N}\cdot\text{m} = 400d \rightarrow d = 2400/400 = 6\text{ m}$. Notice that the LIGHTER person (400 N) needs the LONGER lever arm (6 m) to balance the heavier person's shorter one (3 m) — this can feel backwards if you intuitively expect "more weight needs more distance," but it's actually the opposite: since $\text{moment} = \text{force} \times \text{distance}$, a smaller force needs a proportionally larger distance to produce the same moment.

32. Correct answer: **(B) Equal, opposite, and collinear**

This is a genuinely powerful shortcut, not just a definition to memorize: for a rigid body with exactly two forces acting on it, $\sum M=0$ about EITHER force's own application point forces the OTHER force's line of action to pass through that same point too (since a force has zero moment about any point that lies on its own line of action) — and since this must hold using either point as the reference, both forces must share a single common line of action. Combined with $\sum F=0$ (which needs the two forces to be equal in magnitude, and — since they share a line — opposite in direction), you get the full rule: equal, opposite, and collinear. This single rule is the entire foundation of the Method of Joints in truss analysis (Ch.7): a truss member loaded only at its two end pins IS a two-force member, so its internal force is automatically known to act exactly along the member's own length, with no separate calculation needed to find that direction.

33. Correct answer: **(B) Either concurrent at one point or all parallel**

For a body under exactly three forces, take moments about the intersection point of any two of the three forces — both of those forces then have zero moment about that point (since they pass through it), so the third force's moment about that same point must also be zero for overall equilibrium to hold. But a nonzero force can only have zero moment about a point if its own line of action also passes through that same point. So the third force is forced to pass through the same intersection point as the other two — meaning all three lines of action are concurrent. (The one exception is if the first two forces happen to be parallel and never actually intersect at a finite point — in that special case, the equilibrium condition instead forces all three forces to share the same parallel direction.)

34. Correct answer: **(C) Zero**

A wheel or any ground/roller contact can only PUSH (a positive normal reaction) — it can never pull the object back down, since there's no physical mechanism (like a bolt or clamp) holding the surfaces together. As a vehicle approaches the tipping point about its front wheels, the rear-wheel reaction smoothly shrinks toward zero — that's the exact physical moment of losing contact, right on the edge of lifting off. If your equilibrium math for a stability problem ever produces a NEGATIVE reaction at a simple contact/roller support, that's the mathematical signal that the

assumption of continued contact has broken down (the object has actually tipped), not merely a direction to flip as it would be for a pin or fixed support.

35. Correct answer: **(D) 6**

A general 3D rigid body can potentially translate along any of three independent directions (x, y, z) and rotate about any of three independent axes (x, y, z) — six independent ways it could move if unconstrained. Equilibrium requires blocking ALL six: $\sum F_x=0$, $\sum F_y=0$, $\sum F_z=0$ (no net translation in any direction) plus $\sum M_x=0$, $\sum M_y=0$, $\sum M_z=0$ (no net rotation about any axis). This is the direct 3D generalization of the 2D case, which only needed 3 equations (2 translation directions + 1 rotation axis, since a flat/planar problem only has one meaningful rotation axis, perpendicular to the plane).

Ch. 6 — Center of Gravity & Centroids

36. Correct answer: **(B) $h/3$**

This is one of the most frequently mis-remembered facts in the whole centroid chapter — many students guess $h/2$ by (incorrect) analogy with a rectangle, whose centroid genuinely IS at the geometric midpoint. But a triangle's mass/area is NOT evenly spread with height — there's more area near the wide base than near the narrow tip — so the "balance point" is pulled toward the base, landing at exactly $1/3$ of the height up from the base (equivalently, $2/3$ of the way down from the opposite vertex). This value comes directly from integrating $y \cdot dA$ over the triangle's area and dividing by the total area, but it's worth simply memorizing since it appears constantly in composite-shape problems.

37. Correct answer: **(A) At the intersection of the two lines of symmetry**

If a shape has just one line of symmetry, you know the centroid must lie SOMEWHERE on that line (since the shape balances identically on either side of it) — but you don't yet know exactly where along that line. If the shape has a SECOND, independent line of symmetry, the same logic applies to that line too: the centroid must lie on it as well. A single point can satisfy "lies on line 1" AND "lies on line 2" simultaneously only at the one place those two lines cross — their intersection. This is a pure shortcut (zero computation), and it's why checking for symmetry FIRST, before doing any integration or composite-area arithmetic, is emphasized as the very first step for any centroid problem.

38. Correct answer: **(B) A negative area**

The composite-centroid formula, $\bar{X} = \Sigma(x_i A_i) / \Sigma A_i$, is a weighted average — every piece of the shape contributes its own area (as the "weight") times its own centroid location, all divided by the total area. A hole is literally an ABSENCE of material precisely where solid area would otherwise be, so its contribution must subtract from the total rather than add to it — the mathematically correct way to represent "missing material" in an additive weighted-average formula is to give that region a NEGATIVE area value. Get this sign wrong (treating the hole as ordinary positive

area) and your composite centroid will be pulled the wrong direction, shifted toward the hole instead of away from it.

39. Correct answer: **(C) 0.45 m**

First find the total area by summing the two rectangles: $0.5 + 0.125 = 0.625 \text{ m}^2$. Then apply the weighted-average formula, using each part's own area times its own centroid distance from the shared origin O: $0.625 \cdot \bar{X} = (0.5)(0.5) + (0.125)(0.25) = 0.25 + 0.03125 = 0.28125$. Solving: $\bar{X} = 0.28125/0.625 = 0.45 \text{ m}$. Every distance used here (0.5 m and 0.25 m) was measured from the exact same reference point O — if one part's distance had instead been measured from a different reference, the weighted average would be corrupted; this is exactly why the course stresses using ONE consistent origin for every part of a composite shape.

40. Correct answer: **(B) At the centroid of the triangular load diagram**

When you replace a spread-out (distributed) load with a single equivalent concentrated force, that equivalent force must act at the CENTROID of the load diagram's shape — not at the loaded length's midpoint — because the centroid is precisely the point that captures the "balance" of the entire distribution. For a triangular load, the centroid sits 1/3 of the way from the "thick" (heavily-loaded) end, since more of the load's total weight is concentrated there. Using the midpoint by default (a very tempting shortcut, since it works fine for a RECTANGULAR/uniform load, where the centroid genuinely does coincide with the midpoint) is a common and specifically-flagged mistake for any non-uniform load shape like a triangle or trapezoid.

41. Correct answer: **(B) Zero**

Pore water pressure is defined by $U = \gamma_w \cdot h$, where h is specifically the depth measured below the water table (not below the ground surface, which may sit above the water table). At the water table itself, by definition $h = 0$, so $U = \gamma_w \times 0 = 0$ exactly. Pressure then increases LINEARLY (proportionally) as you go deeper below that reference surface, forming the triangular pressure-distribution diagram used to find the resultant hydrostatic force on a retaining wall or dam. A common mistake is measuring h from the ground surface instead of the water table when the two don't coincide (e.g., the water table lies a few meters below grade) — always identify the water table's own depth first.

Ch. 7 — Structural Analysis

42. Correct answer: **(B) Only at the two end joints**

This is the single defining assumption that makes an entire truss solvable using only simple axial-force methods: every member is treated as a two-force member, meaning it is loaded ONLY at its two pin-connected ends — never along its length, never at any midpoint. Because of that restriction (combined with the two-force-body rule from Ch.5), each member's internal force is guaranteed to act purely along the member's own straight-line axis — pure tension or pure compression, with no bending or shear to worry about. The moment a load gets applied

somewhere OTHER than a joint (mid-span on a member, for instance), that member is no longer a genuine two-force member, and the structure must instead be analyzed as a frame.

43. Correct answer: **(A) Statically determinate**

The truss-determinacy check compares $M+R$ (members plus unknown reactions) against $2J$ (twice the number of joints, since each joint contributes 2 independent equilibrium equations as a 2D particle). Here: $M+R = 9+3 = 12$, and $2J = 2(6) = 12$. Since these are exactly equal, the truss is statically determinate — solvable completely using the Method of Joints or Sections, with neither too few members (which would make it unstable/collapsible) nor too many (which would make it indeterminate, requiring deformation equations beyond this course).

44. Correct answer: **(C) Zero force**

This is a pure-inspection shortcut requiring no algebra at all: at a joint where exactly two non-collinear (not-in-a-straight-line) members meet, and there is no external load or reaction applied directly at that joint, equilibrium in the direction perpendicular to one member forces that member's force to be zero (since nothing else at the joint can balance a force in that direction) — and once that member's force is zero, equilibrium along the OTHER member's direction then forces its force to be zero as well. Scanning an entire truss for this pattern before starting the Method of Joints can eliminate several unknowns immediately, simplifying every remaining joint equation. Note this "zero force" is specific to the CURRENT loading — the member is not necessarily unnecessary for the structure's overall stability or for other load cases.

45. Correct answer: **(B) The force in just one or two specific members**

The Method of Sections lets you "cut" straight through the specific member(s) you care about with an imaginary line, isolate one resulting half of the truss as a free body, and apply the three standard rigid-body equilibrium equations ($\sum F_x=0$, $\sum F_y=0$, $\sum M=0$) directly to that section — solving for up to three unknown member forces (the ones your cut crossed) in one step, without needing to solve every joint between the truss's edge and the member you actually want. This makes it far faster than the Method of Joints specifically when you only need one or two "deep interior" member forces; if you need EVERY member's force, the Method of Joints (marching systematically joint to joint) is the more natural, systematic choice.

46. Correct answer: **(B) At least one multi-force member**

This is the exact defining line between a truss and a frame: a truss requires EVERY member to be a two-force member (loaded only at its two end pins); a frame has at least one member acted on by three or more forces — for example, a member with a load applied somewhere along its length, not just at its two connection points. Because a multi-force member's internal force is no longer guaranteed to act purely along its own axis, frame analysis requires the more general "dismember and draw a separate FBD for each individual piece" approach, rather than the streamlined joint/section shortcuts that work specifically because of the two-force-member assumption in trusses.

47. Correct answer: **(C) Equal in magnitude, opposite direction (same line of action)**

This is a direct application of Newton's Third Law at the shared pin: whatever force member 1 exerts on member 2 at that connection, member 2 exerts the exact equal-and-opposite force back on member 1, along the same line of action. In practice, this means once you solve for the pin force using ONE member's free body, you get the OTHER member's pin force for free — just flip the sign — rather than having to derive it independently from scratch (though independently re-deriving it IS a valuable check, since agreement between the two confirms your solution is self-consistent).

48. Correct answer: **(B) Amplifies the output force above the input**

Taking moments about the pivot for a machine like pliers gives $Pb = Qa$ (the input force P times its own lever arm b must balance the output force Q times its own lever arm a), which rearranges to $Q = P(b/a)$. If the input force is applied FARTHER from the pivot than the output contact point ($b > a$), then the ratio b/a is greater than 1, making $Q > P$ — the tool amplifies force. This is exactly how pliers, bolt cutters, and similar hand tools achieve large gripping or cutting forces from a comparatively modest hand squeeze: a long handle (large b) paired with a short jaw distance from the pivot (small a).

Ch. 8 — Internal Forces, Stress & Strain

49. Correct answer: **(B) 100 lb/in²**

Stress is defined simply as Force divided by the cross-sectional Area over which that force acts: $\text{Stress} = 1000 \text{ lb} / 10 \text{ in}^2 = 100 \text{ lb/in}^2$ (also written as 100 psi, "pounds per square inch"). Stress matters more than raw force alone because the SAME force can be harmless or catastrophic depending purely on how concentrated it is — a knife cuts because it focuses force onto a tiny edge area (huge stress), while a flat palm applying the same force is harmless (the force is spread over a much larger area, so the stress is much lower). Material failure limits (like "steel yields at 250 MPa") are always stated as stress values, never as raw force, precisely because the same material can safely carry very different absolute forces depending on its cross-sectional size.

50. Correct answer: **(B) 0.012 (dimensionless)**

Strain is defined as the change in length divided by the ORIGINAL length: $\text{Strain} = \Delta L / L_0 = 0.12 \text{ in.} / 10 \text{ in.} = 0.012$. Because both the numerator and denominator are measured in the same unit (inches, in this case), those units cancel out completely in the division — strain is a pure, dimensionless number, meaning its numeric value would be exactly the same 0.012 even if you'd measured the original bar in centimeters or meters instead of inches, as long as you used the SAME unit consistently for both ΔL and L_0 . This dimensionless-ness is exactly what makes strain useful for comparing how "stretchy" objects of very different sizes are on a fair, apples-to-apples basis.

51. Correct answer: **(C) Flat (constant)**

The shear diagram's slope at any point along a beam equals the negative of the distributed load intensity there: $dV/dx = -w(x)$. Where there is NO distributed load acting ($w=0$ along that stretch of the beam), the shear diagram's slope is exactly zero — meaning V doesn't change at all along that segment; it stays perfectly flat (horizontal on the diagram) until the next point load causes a sudden jump, or a distributed load begins and introduces a nonzero slope again. Recognizing these slope-shape relationships lets you sketch an entire shear (and moment) diagram quickly by eye, without recomputing V and M by direct cutting at every single point along the beam.

52. Correct answer: **(B) The shear diagram crosses zero**

The moment diagram's slope at any point equals the shear value there: $dM/dx = V$. A function's slope is zero exactly at its local maxima and minima — so the moment diagram has a horizontal tangent (a peak or trough, meaning a candidate for the beam's overall maximum bending moment) specifically at the point(s) where the shear diagram crosses through zero. This is the standard, reliable technique engineers use to locate the critical design moment along a beam — simply looking for where the largest applied LOAD is (a tempting but often wrong shortcut) does not reliably locate the true maximum moment, since moment depends on the accumulated effect of shear over the beam's length, not just the size of any one load.

Ch. 9 — Friction

53. Correct answer: **(B) The normal force pressing the surfaces together**

The friction formula $F_f = \mu F_N$ shows friction force scaling directly and only with the normal force F_N (the force pressing the two surfaces together) and the friction coefficient μ (a property of which two materials are in contact) — there is no area term in this formula at all. This runs against a very common intuition: many people assume a wider tire or a bigger block "grips more" because it has more contact area, but in the simple Coulomb (dry) friction model used throughout this course, friction is explicitly independent of contact area; only the normal force and the material pairing matter.

54. Correct answer: **(C) $F_s > F_k$**

For essentially every real material pair, the maximum available static friction force is larger than the kinetic (sliding) friction force between the same two surfaces — this matches everyday experience directly: pushing a heavy couch across a floor takes a large initial shove to overcome static friction and get it moving, but once it's sliding, a noticeably smaller continuous push is enough to keep it moving at constant speed, since kinetic friction is weaker. This is exactly why the coefficient tables in this chapter list two separate values for every material pair — μ_s (static) is always listed higher than μ_k (kinetic) for the same two materials.

55. Correct answer: **(B) 363 N**

Maximum static friction is found the same way as any friction force, using the STATIC coefficient specifically: $F_{f,max} = \mu_s \times F_N = 0.74 \times 490 \text{ N} = 362.6 \text{ N}$, which rounds to 363 N.

This is the largest friction force the contact can supply before sliding begins — any applied horizontal force smaller than 363 N will be exactly matched by static friction (keeping the block stationary); the moment the applied force tries to exceed 363 N, the block breaks free and starts sliding, at which point friction actually DROPS to the smaller kinetic value ($0.57 \times 490 \approx 279$ N, close to distractor (A) 280 N — that distractor exists to test whether you correctly used μ_s rather than accidentally grabbing μ_k).

56. Correct answer: **(C) The maximum static friction force, F_m**

"Motion impending" is a precise technical phrase describing the exact instant right before sliding actually begins: the object is still at rest, but the applied force P has grown to exactly match the LARGEST friction force the surfaces can supply, $F_m = \mu_s F_N$. Any further increase in P and the object starts to slide. This phrase is a strong signal in problem statements: it tells you to use the maximum static friction formula ($F_f = \mu_s F_N$, at its full/maximum value) rather than treating friction as some smaller, not-yet-maximum value, and rather than switching to the kinetic formula (which only applies once actual sliding has begun).

57. Correct answer: **(C) Is treated as roughly constant, independent of speed**

The Coulomb (dry) friction model that this entire chapter is built on makes a deliberate simplifying assumption: once an object is sliding, the kinetic friction coefficient μ_k is treated as one single constant number, regardless of how fast or slow the object happens to be moving at any given instant. This is what makes kinetic-friction problems solvable with a single, unchanging μ_k value throughout a calculation — for example, a hockey puck decelerating across ice experiences the same μ_k right after being struck (moving fast) as it does near the very end of its slide (moving slowly), in this idealized model. It's worth remembering this is a MODELING assumption, not an absolute physical law — real friction can show some speed-dependence in certain regimes — but every problem in this course consistently uses the constant- μ_k simplification.